

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – SCENARIO BASED

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 2 – SCENARIO BASED
Weightage:	30% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	CSE-AI	Date:	1-08-2026

Code of Conduct

I N Goutham(192472024)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N Goutham

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Tagsets for English, Part of Speech Tagging	Morphological decomposition of analyzing, analysis, and analytical; identification of roots, affixes, inflectional and derivational forms, and normalization for indexing.	Q1
English Word Classes, Tagsets for English, Stochastic Part of Speech Tagging	Morphological parsing of disagree, agreement, and agreeable; identification of prefixes, suffixes, base forms, and semantic effects of derivation.	Q2
Counting Words, English Word Classes, Part of Speech Tagging	Morphology-based normalization of govern, government, and governance for topic modeling, clustering, and lexical standardization.	Q3
Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Morphological analysis of activate, activation, and reactivation; identification of derivational layers, word-class changes, and semantic shifts.	Q4
Rule-Based Part of Speech Tagging, English Word Classes, Part of Speech Tagging	Inflectional normalization of create, creates, and creating; extraction of root forms and classification of tense/aspect transformations.	Q5

Annexure A – SCENARIO BASED

1. A large-scale search engine indexing system processes user queries containing word variants such as “analyzing”, “analysis”, and “analytical”. The system must improve retrieval accuracy by grouping semantically related terms under a unified representation. Design a morphological processing approach to decompose each input word into root and affixes, distinguish between inflectional and derivational forms, and normalize them for indexing. Apply the process to the given inputs and determine the root form along with the morphological structure of each word.

Q1. Morphological Processing for Search Engine Indexing

Scenario Understanding

A large-scale search engine receives user queries containing different forms of the same word such as **"analyzing," "analysis," and "analytical."** If these words are stored separately in the index, users may miss relevant documents. Therefore, the system requires a morphological processing approach to identify their common root and normalize them into a unified representation, improving indexing efficiency and retrieval accuracy.

Key Issues Identified

- Different word forms represent the same underlying concept.
- Inflectional and derivational forms must be distinguished.
- Multiple word variants should be indexed under one common representation.
- Normalization should improve search accuracy without losing semantic meaning.

Morphological Processing Strategy

The proposed morphological processing pipeline consists of the following steps:

Step 1: Input Processing

- Receive the words: **analyzing, analysis, analytical**

Step 2: Morphological Decomposition

- Separate each word into root and affixes.

Step 3: Morphological Classification

- Determine whether the transformation is **inflectional** or **derivational**.

Step 4: Normalization

- Convert all variants into a common base form for indexing.

Morphological Analysis

Word 1: analyzing

- Root : **analyze**
- Suffix : **-ing**
- Word Class : Verb
- Morphological Type : **Inflectional**

- Explanation:
The suffix **-ing** represents the present participle or continuous tense. It changes only the grammatical form without changing the meaning.

Word 2: analysis

- Root : **analyze**
- Suffix : **-sis**
- Word Class : Noun
- Morphological Type : **Derivational**
- Explanation:
The suffix changes the verb into a noun, representing the process or result of analyzing.

Word 3: analytical

- Root : **analyze**
- Suffix : **-ical**
- Word Class : Adjective
- Morphological Type : **Derivational**
- Explanation:
The suffix converts the base concept into an adjective describing something related to analysis.

Decision and Justification

The search engine should normalize all three words to the common base form "**analyze**." Although the words belong to different grammatical categories, they share the same semantic meaning. Indexing them under a unified representation reduces duplicate index entries, improves recall, and retrieves all relevant documents for a user query.

Final Morphological Output

Input Word	Root	Affix	Type	Normalized Form
analyzing	analyze	-ing	Inflectional	analyze
analysis	analyze	-sis	Derivational	analyze
analytical	analyze	-ical	Derivational	analyze

Conclusion

The proposed morphological processing approach successfully decomposes each word into its root and affixes, distinguishes inflectional and derivational forms, and normalizes all variants to the common root "**analyze**." This normalization improves indexing efficiency, reduces redundancy, and enhances search retrieval performance by ensuring that semantically related words are grouped together.

2. An AI-based sentiment analysis platform receives product reviews containing words like “disagree”, “agreement”, and “agreeable”. The system must interpret sentiment consistently despite variations in word formation. Develop a morphological parsing strategy to identify prefixes, suffixes, and base forms, while capturing how derivational changes influence meaning. Process the given inputs to extract roots and classify each transformation, ensuring accurate semantic interpretation across all word forms.

Q2. Morphological Parsing for AI-Based Sentiment Analysis

Scenario Understanding

An AI-based sentiment analysis platform processes customer reviews containing words such as **"disagree," "agreement," and "agreeable."** These words originate from the same base word but differ in meaning due to prefixes and suffixes. If the system treats them as unrelated words, it may misinterpret the sentiment expressed in the review. Therefore, a morphological parsing strategy is required to identify prefixes, suffixes, and root forms while preserving semantic consistency across all word variants.

Key Issues Identified

- Different word forms share the same base word **"agree."**
- Prefixes and suffixes modify the meaning and grammatical category.
- The system must distinguish derivational transformations from the base form.
- All variants should be linked to a common root for accurate sentiment interpretation.

Morphological Parsing Strategy

The proposed morphological parser performs the following steps:

Step 1: Input Processing

Receive the input words:

- disagree
- agreement
- agreeable

Step 2: Morphological Decomposition

Separate each word into:

- Prefix (if present)
- Root word
- Suffix (if present)

Step 3: Morphological Classification

Determine whether each transformation is derivational or inflectional.

Step 4: Semantic Interpretation

Analyze how prefixes and suffixes modify the meaning while preserving the common root.

Step 5: Normalization

Normalize all words to the common base form **"agree"** for consistent sentiment analysis.

Morphological Analysis

Word 1: disagree

- Prefix : **dis-**
- Root : **agree**
- Suffix : —
- Word Class : Verb
- Morphological Type : **Derivational**

Explanation:

The prefix **dis-** changes the meaning from positive agreement to negative disagreement. The grammatical category remains a verb, but the semantic meaning becomes opposite.

Word 2: agreement

- Prefix : —
- Root : **agree**
- Suffix : **-ment**
- Word Class : Noun
- Morphological Type : **Derivational**

Explanation:

The suffix **-ment** converts the verb into a noun representing the result or state of agreeing.

Word 3: agreeable

- Prefix : —
- Root : **agree**
- Suffix : **-able**
- Word Class : Adjective
- Morphological Type : **Derivational**

Explanation:

The suffix **-able** changes the verb into an adjective describing a person or thing that is pleasant or willing to agree.

Decision and Justification

The sentiment analysis system should normalize all three words to the common root "**agree**." During sentiment interpretation, the system should also preserve the semantic effect of the prefix and suffix.

- **disagree** → Negative sentiment because of the prefix **dis-**.
- **agreement** → Neutral or positive depending on context.
- **agreeable** → Positive sentiment because it describes a pleasant quality.

This approach ensures accurate sentiment classification while maintaining semantic relationships among related word forms.

Final Morphological Output

Input Word	Prefix	Root	Suffix	Type	Normalized Form
disagree	dis-	agree	—	Derivational	agree
agreement	—	agree	-ment	Derivational	agree
agreeable	—	agree	-able	Derivational	agree

Conclusion

The proposed morphological parsing strategy successfully identifies prefixes, suffixes, and root forms while preserving semantic consistency. By normalizing all variants to the common root "**agree**" and considering the semantic impact of derivational morphology, the sentiment analysis platform can accurately interpret positive, negative, and neutral expressions in user reviews. This improves the overall accuracy and reliability of sentiment classification.

3. A text analytics pipeline in a news aggregation system handles documents containing terms such as "govern", "government", and "governance". The system must standardize these variations for topic modeling and clustering. Construct a morphology-based normalization mechanism that decomposes each word into its root and affixes, identifies derivational layers, and maps them to a common base form. Apply the approach to the input words and derive the normalized representation along with their morphological breakdown.

Q3. Morphology-Based Normalization for Topic Modeling and Clustering Scenario Understanding

A news aggregation system processes articles containing words such as "**govern**," "**government**," and "**governance**." These words have the same lexical origin but appear in different forms. If they are treated as separate terms, the topic modeling and clustering process may generate inaccurate results. Therefore, the system requires a morphology-based normalization mechanism to identify the common root, remove derivational affixes, and map all variants to a unified representation for consistent document analysis.

Key Issues Identified

- The words **govern**, **government**, and **governance** originate from the same root.
- Suffixes change the grammatical category and semantic meaning.
- Multiple word variants should be normalized to a single base form.
- Uniform normalization improves topic modeling, document clustering, and lexical consistency.

Morphology-Based Normalization Strategy

The proposed normalization mechanism consists of the following steps:

Step 1: Input Processing

Receive the input words:

- govern
- government

- governance

Step 2: Morphological Decomposition

Separate each word into:

- Root
- Suffix

Step 3: Derivational Analysis

Identify how each suffix changes the grammatical category and meaning.

Step 4: Normalization

Map all word variants to the common base form "**govern.**"

Step 5: Indexing

Store the normalized root for topic modeling and document clustering to improve lexical consistency.

Morphological Analysis

Word 1: govern

- Root : **govern**
- Suffix : —
- Word Class : Verb
- Morphological Type : Base Form

Explanation:

The word **govern** is the original verb representing the action of controlling or administering. It serves as the common root for all related forms.

Word 2: government

- Root : **govern**
- Suffix : **-ment**
- Word Class : Noun
- Morphological Type : **Derivational**

Explanation:

The suffix **-ment** converts the verb into a noun referring to the governing authority or institution.

Word 3: governance

- Root : **govern**
- Suffix : **-ance**
- Word Class : Noun
- Morphological Type : **Derivational**

Explanation:

The suffix **-ance** transforms the verb into a noun representing the process, practice, or system of governing.

Decision and Justification

The news aggregation system should normalize all three words to the common base form **"govern."** Although **government** and **governance** are different noun forms, they are semantically related to the same governing concept. Storing them under a unified root reduces vocabulary size, improves document clustering, and increases the accuracy of topic modeling by grouping related documents together.

Final Morphological Output

Input Word	Root	Suffix	Word Class	Morphological Type	Normalized Form
govern	govern	—	Verb	Base Form	govern
government	govern	-ment	Noun	Derivational	govern
governance	govern	-ance	Noun	Derivational	govern

Conclusion

The proposed morphology-based normalization mechanism successfully decomposes each word into its root and suffix, identifies derivational transformations, and maps all related forms to the common base word **"govern."** This normalization improves lexical consistency, reduces redundant vocabulary, and enhances the performance of topic modeling and document clustering by grouping semantically related terms under a unified representation.

4.A document classification system processes input words such as “activate”, “activation”, and “reactivation” to improve semantic grouping and indexing.

Construct a morphological parsing strategy to identify prefixes, suffixes, and root forms while distinguishing derivational layers.

Analyze how prefix addition and suffix transformation affect meaning and word class.

Apply the process to decompose each word and map them to a unified base representation.

Derive the morphological structure and normalized root for each input word.

Q4. Morphological Parsing for Document Classification

Scenario Understanding

A document classification system processes words such as **"activate," "activation," and "reactivation."** Although these words are morphologically related, they differ in grammatical category and meaning due to the addition of prefixes and suffixes. If the system treats them as separate terms, related documents may be classified into different categories. Therefore, a morphological parsing strategy is required to identify prefixes, suffixes, and root forms, distinguish derivational layers, and normalize all variants to a unified base representation for accurate document classification and semantic grouping.

Key Issues Identified

- The words **activate, activation, and reactivation** share the same root **"activate."**
- Prefixes and suffixes modify both meaning and word class.

- The system must identify multiple derivational layers.
- All related forms should be mapped to a common base representation for consistent indexing and classification.

Morphological Parsing Strategy

The proposed morphological parsing strategy consists of the following steps:

Step 1: Input Processing

Receive the input words:

- activate
- activation
- reactivation

Step 2: Morphological Decomposition

Separate each word into:

- Prefix (if present)
- Root
- Suffix

Step 3: Derivational Layer Analysis

Identify how each prefix and suffix changes the grammatical category and semantic meaning.

Step 4: Normalization

Map all variants to the common base form "**activate**."

Step 5: Classification

Store the normalized form for document classification and semantic grouping.

Morphological Analysis

Word 1: activate

- Prefix : —
- Root : **activate**
- Suffix : —
- Word Class : Verb
- Morphological Type : Base Form

Explanation:

The word **activate** is the original verb meaning *to make something active*. It serves as the common base form for all derived words.

Word 2: activation

- Prefix : —
- Root : **activate**
- Suffix : **-ion**
- Word Class : Noun
- Morphological Type : **Derivational**

Explanation:

The suffix **-ion** converts the verb into a noun representing the process or result of activating. The word class changes from **verb** to **noun**, while the core meaning remains related.

Word 3: reactivation

- Prefix : **re-**
- Root : **activate**
- Suffix : **-ion**
- Word Class : Noun
- Morphological Type : **Derivational**

Explanation:

The prefix **re-** adds the meaning "**again**", while the suffix **-ion** converts the verb into a noun. Thus, **reactivation** means **the process of activating again**.

Decision and Justification

The document classification system should normalize **activate**, **activation**, and **reactivation** to the common base form "**activate**." Although the derived forms belong to different grammatical categories and include additional semantic information, they represent the same core concept. Mapping them to a unified representation reduces redundancy, improves semantic grouping, and increases the accuracy of document classification and indexing.

Final Morphological Output

Input Word	Prefix	Root	Suffix	Word Class	Morphological Type	Normalized Form
activate	—	activate	—	Verb	Base Form	activate
activation	—	activate	-ion	Noun	Derivational	activate
reactivation	re-	activate	-ion	Noun	Derivational	activate

Conclusion

The proposed morphological parsing strategy successfully identifies prefixes, suffixes, and root forms while distinguishing derivational layers. By normalizing **activate**, **activation**, and **reactivation** to the common base form "**activate**," the document classification system achieves better semantic grouping, consistent indexing, and improved classification accuracy. This approach also preserves the semantic effects of prefix addition and suffix transformation while maintaining a unified lexical representation.

5. A search optimization module processes input words such as “create”, “creates”, and “creating” to ensure consistent retrieval across grammatical variations. Design a morphology-based normalization approach focusing on inflectional transformations such as tense and aspect.

Identify suffix patterns and map all variations to a common base form without altering meaning.

Apply the process to extract root forms and classify each transformation.

Determine the morphological breakdown and normalized output for each word.

Q5. Morphology-Based Normalization for Search Optimization

Scenario Understanding

A search optimization module processes words such as **"create," "creates," and "creating."** These words represent different grammatical forms of the same verb. If they are indexed separately, users may not retrieve all relevant documents when searching for a single term. Therefore, the system requires a morphology-based normalization approach to identify inflectional transformations, extract the common root, and normalize all variants without changing their semantic meaning.

Key Issues Identified

- The words **create, creates, and creating** share the same root **"create."**
- The suffixes represent grammatical changes such as tense and aspect.
- These transformations are **inflectional** because they do not alter the core meaning.
- All word forms should be normalized to a common base form for efficient search and retrieval.

Morphology-Based Normalization Strategy

The proposed normalization approach consists of the following steps:

Step 1: Input Processing

Receive the input words:

- create
- creates
- creating

Step 2: Morphological Decomposition

Separate each word into:

- Root
- Suffix

Step 3: Inflectional Analysis

Identify whether the suffix represents tense or aspect without changing the lexical meaning.

Step 4: Normalization

Map all word forms to the common base form **"create."**

Morphological Analysis

Word 1: create

- Root : **create**
- Suffix : —
- Word Class : Verb
- Morphological Type : **Base Form**

Explanation:

The word **create** is the original verb and serves as the common root for all grammatical variations.

Word 2: creates

- Root : **create**
- Suffix : **-s**
- Word Class : Verb
- Morphological Type : **Inflectional**

Explanation:

The suffix **-s** indicates the **third-person singular present tense**. It changes only the grammatical form while preserving the original meaning.

Word 3: creating

- Root : **create**
- Suffix : **-ing**
- Word Class : Verb
- Morphological Type : **Inflectional**

Explanation:

The suffix **-ing** represents the **present participle or continuous aspect**. It expresses an ongoing action without changing the core meaning of the verb.

Decision and Justification

The search optimization module should normalize **create**, **creates**, and **creating** to the common base form "**create**." Since the suffixes **-s** and **-ing** represent only grammatical variations, storing separate index entries would increase redundancy. Normalizing all forms to the root ensures that users retrieve all relevant documents regardless of the grammatical form used in the query, thereby improving search efficiency and retrieval accuracy.

Final Morphological Output

Input Word	Root	Suffix	Transformation	Morphological Type	Normalized Form
create	create	—	Base Form	Base Form	create
creates	create	-s	Third-person singular present tense	Inflectional	create
creating	create	-ing	Present participle / Continuous aspect	Inflectional	create

Conclusion

The proposed morphology-based normalization approach successfully identifies the root and inflectional suffixes of **create**, **creates**, and **creating**. By mapping all grammatical variations to the common base form "**create**," the search optimization module reduces redundancy, improves indexing efficiency, and ensures consistent retrieval of relevant documents. This approach enhances search performance while preserving the original semantic meaning of the words.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Q. No. / Criteria Level	Understanding of Scenario	Identification of Key Issues	Application of Concepts	Decision Making	Clarity of Response	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____